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Lab	04
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Department of Software Engineering	

LAB MANUAL: STAR TOPOLOGY USING SWITCH

TITLE:

Network Communication Using Switch (Packet Tracer)

Objective:

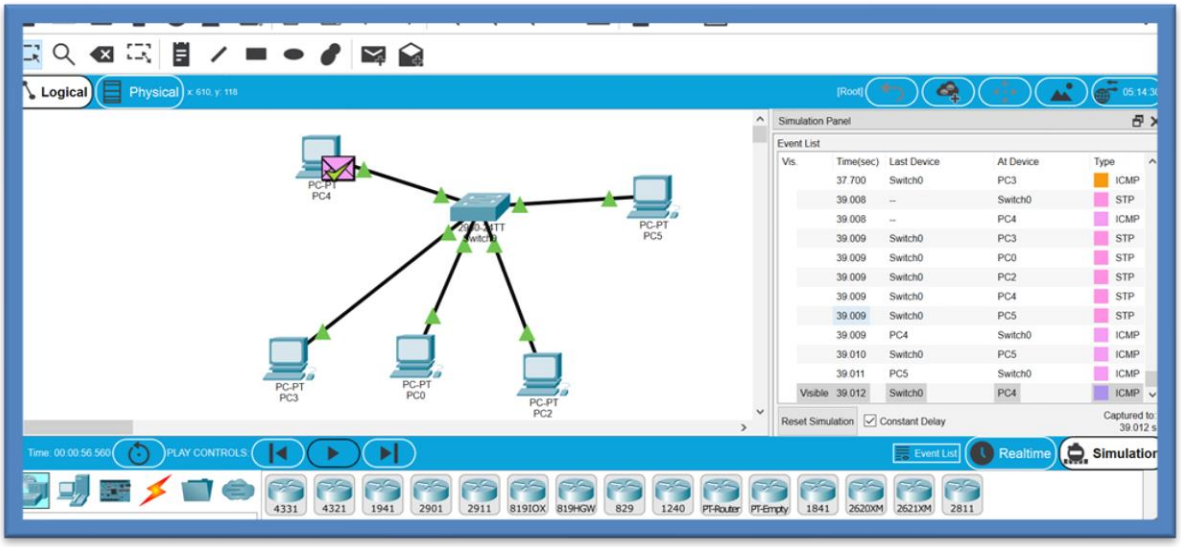
To learn how multiple PCs communicate through a switch and observe packet flow.

Main Concepts:

Switch: Connects multiple devices and forwards data using MAC address.
IP Address: Unique identifier for each computer (e.g., 192.168.1.1).
MAC Address: Physical address used by switch to send data.
ICMP (Ping): Checks connectivity between devices.
STP: Prevents loops in the network.
Simulation Mode: Shows packet movement step-by-step.

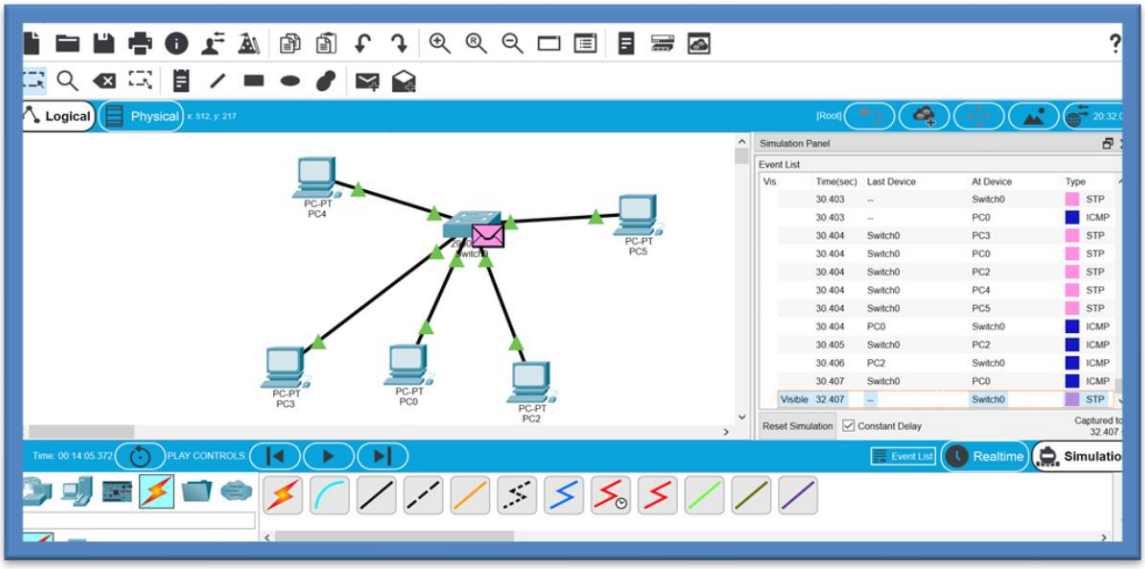
Procedure:

1. Add 1 switch and 5 PCs
2. Connect all PCs to the switch
3. Assign IP addresses
4. Use ping command
5. Switch to Simulation Mode
6. Observe packet flow



Result:

All PCs communicated successfully through the switch.



CONCLUSION:

Switch enables efficient communication in LAN and simulation helps visualize data flow.